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Prediction of Consultation Wait Time in Outpatient Clinic: An Approach using Neural Network with Optimized Feature Selection

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Abstract

Outpatient clinics globally grapple with the uncertainty of patient wait times, a critical factor affecting patient satisfaction. Extended waiting periods are often perceived as a hindrance to timely care, generating stress for both patients and healthcare providers. Accurate prediction of patient wait times can substantially improve patient satisfaction by reducing uncertainty. This study aims to predict patient Consultation Wait Time efficiently in a multispecialty hospital outpatient clinic utilizing a Multilayer Perceptron approach. Feature Selection Methods were employed to enhance the predictive efficiency of the model. The study evaluated various performance metrics, including Accuracy, Recall, Precision, F-measure, and Area Under the Receiver Operating Characteristic Curve (AUC). Temporal features, specifically Visit Time and Consultation Start Time, emerged as significant predictors in the model. Additionally, Vitals Examination was identified as a key factor in predicting Consultation Wait Time. Notably, the model incorporating variables selected through Reciprocal Ranking exhibited robust performance in predicting Consultation Wait Times.

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Keywords: Machine Learning; Wait Time; Predictive Analytics; Feature Selection; Reciprocal Ranking; Neural Network; ROC;

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1. Introduction

Prolonged waiting times in outpatient clinics have garnered significant academic and clinical attention as a critical concern in global healthcare delivery systems. The link between extended waiting times and patient satisfaction has been extensively documented. McCarthy et al. (2000) underscored the importance of healthcare institutions prioritizing efficient wait time management to deliver high-quality service [1]. This sentiment is reinforced by studies conducted by Nottingham et al. (2018), Donahue et al. (2017), Ogaji & Mezie-Okoye (2017), and Alarcon-Ruiz et al. (2019), collectively establishing a clear negative relationship between prolonged wait times and patient satisfaction in outpatient settings [2, 3, 4, 5].

Accurate prediction of waiting times is not merely a daily operational requirement for clinics; it is a strategic necessity that significantly affects a clinic's credibility and overall reputation. The present study aims to contribute to this discussion by employing a Multilayer Perceptron Model (MLP) to predict Consultation Wait Time, which refers to the duration patients spend waiting to consult with a healthcare professional. The selection of a Multilayer Perceptron Model (MLP) in this study is based on several key factors, including its robustness, adaptability, and a proven track record in similar predictive tasks, making it particularly suitable for predicting Consultation Wait Time in healthcare settings [6].

The problem of elongated waiting times is universal, yet its manifestations can be context-dependent. For instance, a study from Nigeria reported that a substantial proportion of patients had consultations lasting less than 5 minutes, yet 61% endured waiting periods ranging between 90 and 180 minutes [7]. Sector-specific investigations have been conducted across diverse healthcare departments, such as radiology, maternity, and emergency services emphasizing that priority status, current service status, and queue size serve as key determinants in outpatient wait time predictions [8,9,10,11]. Other influencing factors like educational level, place of residence, day of the visit, and arrival time were explored [12]. Various methodologies, including logistic regression and machine learning techniques like neural networks, random forests, and support vector machines, have been utilized in previous studies [10,13,14]. In a study, Pianykh (2015) employed neural network methodologies to predict patient wait times in healthcare settings, achieving notable accuracy with a prediction error as low as 31% for L1 regression, demonstrating the efficacy of neural networks in predicting wait times in complex clinical environments [15].In another targeted research on phlebotomy units, Javadifard et al. successfully applied a neural network algorithm, achieving an impressive 88% accuracy in predicting patient waiting times, thereby underscoring the potential of neural networks in effectively forecasting wait times[16].In another research, Kuo et al. demonstrated that machine learning algorithms, particularly artificial neural networks, significantly outperformed traditional linear regression models in predicting waiting times in emergency departments, showcasing a reduction in mean-square error by approximately 20%[17].

An MLP classifier is employed in the current study to predict Consultation Wait Times in outpatient healthcare settings, with specific attention given to factors like temporal factors and patient interaction factors. In contrast, the studies cited examine various healthcare contexts using different machine learning models to predict wait times in settings such as emergency departments and specialized units like phlebotomy. The current study considered predicting Consultation Wait Time, as a multiclass classification problem which also adds to the novelty. Another notable feature of this research is the introduction of a novel metric, the Consultation Wait Time Median of 5 days (Cwtm5), as a crucial measure for predictive modeling. Importantly, the absence of similar studies employing the exact set of variables used in this study shows the innovative nature of this research. Additionally, various feature selection techniques, including Pearson Correlation, Recursive Feature Elimination, Random Forest Classifier, and Reciprocal Ranking, have been strategically applied to optimize the performance of the MLP model in predicting consultation wait times [18]. These techniques fall into different categories, including wrapper, embedded, hybrid, and filter types, each offering its unique advantages and limitations [19]. This methodological diversity further contributes to the novelty and depth of our study.

The performance of the models is measured using Accuracy, Recall, Precision, F-measure, and Area under the Receiver Operating Characteristic Curve. The subsequent sections of this paper are organized as follows: Section 2 delineates the methodology and development of the prediction model, Section 3 explicates the feature selection techniques employed, and Section 4 assesses the model's performance metrics. Discussions of the findings are found in Section 5, limitations in Section 6, and the paper concludes in Section 7.

2. Model Development

2.1 Framework of the study

The data used for the study was collected from the electronic health records of an outpatient department of a multispeciality hospital in India for six months. The cleaned and preprocessed data consisted of 5600 patient records. To enhance the prediction of wait times, new variables are extracted through feature engineering. All the variables considered for the prediction of Consultation wait times are Gender, Visit Day, Visit Month, Visit Time, Priority, Consulting Session, Vitals Examination, Consultation Start Time, Repeat Arrival, and Consultation Wait Time Median of 5 days (Cwtm5). Various feature selection techniques are employed to select the most suitable features for predicting Consultation Wait Time. Five distinct models were formulated, each diverging in the set of predictor variables incorporated. Model 1 encompassed all available variables, thereby serving as a comprehensive baseline. Model 2 was predicated on variables filtered through Pearson Correlation, offering a statistical approach to feature selection. Model 3 leveraged Recursive Feature Elimination (RFE) as its selection algorithm, for determining the features. Model 4 incorporated variables selected by a Random Forest Classifier, adding an ensemble learning method to find the feature set. Finally, Model 5 utilized the feature selection approach of Reciprocal Ranking (RR), providing yet another angle for the variable selection. These feature selection techniques fall into different categories, including wrapper, embedded, hybrid, and filter types, each offering its unique advantages and limitations. This methodological diversity contributes to the novelty and depth of the study. These multiple models offer a comparative framework for assessing the impact of different variable sets on the predictive accuracy of Consultation Wait Times. The models are trained using 85% of the data and then tested using 15% of the data. The study used the Scikit-learn library in Python to implement the Multilayer Perceptron Classifier to predict the wait times. The performances of all the models are measured using performance indicators such as Accuracy, Precision, Sensitivity, F measure, and Area under the Receiver Operating Characteristic Curve.

Thus, the study focused on the examination of the efficiency of predicting this wait time utilizing a Multilayer Perceptron model and the identification of factors contributing to patient Consultation Wait Time in a multispecialty hospital outpatient clinic. The hypothesis posited that significant predictors of Consultation Wait Time would include certain temporal features, specifically Visit Time, Consulting Session, and Consultation Start Time. Furthermore, it was anticipated that strong predictive performance in predicting Consultation Wait Time would be demonstrated by the Multilayer Perceptron model when trained with feature selection methods like Reciprocal Ranking. Fig.1. depicts the study's overall framework.

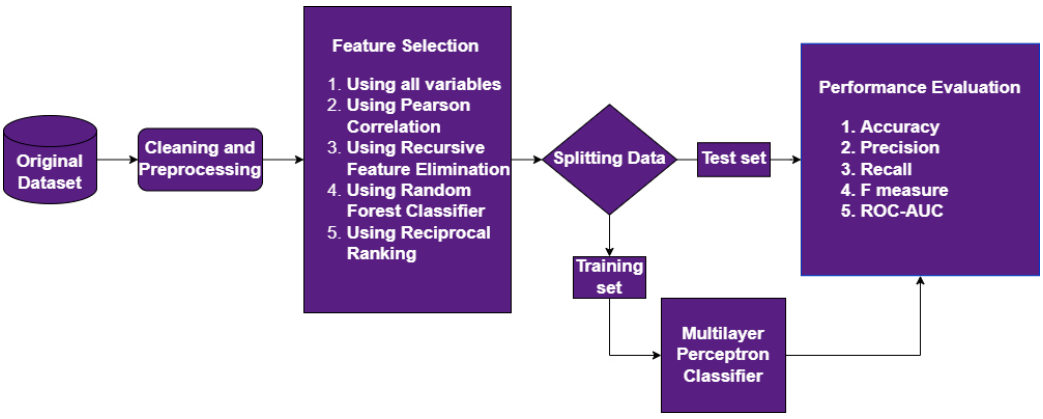


Fig. 1. Multilayer Perceptron model for predicting Consultation Wait Time

2.2. Analysis of Variables

In the baseline model of this study, a multitude of predictor variables Gender, Visit Time, Visit Day, Visit Month, Consultation Start Time, Repeat Arrival, Priority, Vitals Examination, Consulting Session, and Consultation Wait Time Median of Five Days (Cwtm5) were utilized to predict Consultation Wait Time. The latter serves as the dependent variable in the study and is categorized into three distinct levels: low, medium, and high, thereby constituting a multiclass classification problem. The operational definitions for these categories are as follows: "low" refers to wait durations that are less than 30 minutes, "medium" encompasses the 30-to-60-minute range, and "high" is assigned to durations that extend beyond 60 minutes. A comprehensive analysis of the predictor variables' impact on Consultation Wait Time is consolidated in Table 1.

Table 1. Analysis of variables

Features		**Consultation Wait time**			***Total Number of Patients***
		L	*M*	*H*	
Gender	Female	2138	1048	154	3340
	Male	1558	608	94	2260
Visit Day	Mon	782	340	54	1176
	Tue	564	236	62	862
	Wed	584	248	28	860
	Thru	612	226	26	864
	Fri	520	300	46	866
	Sat	578	278	24	880
	Sun	56	28	8	92
Visit Month	Feb	208	186	30	424
	March	592	260	48	900
	April	656	346	50	1052
	May	854	338	30	1222
	June	670	294	54	1018
	July	716	232	36	984
Visit Time	8-9 am	0	18	10	28
	9-10 am	236	272	64	572
	10-11 am	982	534	56	1572
	11-12 pm	1092	350	40	1482
	12 pm-1 pm	562	126	2	690
	1 pm- 2 pm	86	2	0	88
	2 pm -3 pm	0	0	0	0
	3 pm -4 pm	0	0	10	10
	4 pm -5 pm	42	182	40	264
	5 pm -6 pm	358	138	22	518
	6 pm - 7 pm	292	34	2	328
	7 pm- 8 pm	46	0	2	48
Priority	Low	2540	946	248	3734
	High	1156	710	0	1866
Vitals Examination	No	2314	942	144	3400
	Yes	1382	714	104	2200
Consulting Session	Morning	2958	1302	172	4432
	Evening	738	354	76	1168

Consultation Start Time	Before 10 am	1538	630	72	2240
	10 -10.15 am	512	180	16	708
	10.15 -10.30 am	596	284	28	908
	10.30 - 10.45 am	246	182	40	468
	10.45 am-11am	64	24	4	92
	After 11 am	2	2	12	16
	4 pm -4.15 pm	214	82	14	310
	4.15 pm to 4.30 pm	314	190	38	542
	4.30 pm to 4.45 pm	134	60	22	216
	4.45 pm to 5 pm	38	20	2	60
	After 4 pm	38	2	0	40
Repeat Arrival	Yes	1184	528	90	1802
	No	2512	1128	158	3798
Consultation Wait Time Median 5 days (Cwtm5)	< 20 min	472	210	48	730
	20-30 min	2612	1118	152	3882
	31-40 min	548	318	42	908
	> 40 min	64	10	6	80

The variable "Repeat Arrival" serves as a variable that distinguishes between initial and recurring visits within the timeframe of the study. The variable "Consultation Wait Time Median of Five Days" (Cwtm5) is calculated based on historical wait time data and represents the median consultation wait time over the preceding five-day period. The feature "Priority" is also incorporated into the analysis, signifying whether the consultation has been flagged as a priority appointment. The temporal aspects of the consultation are captured through two additional variables: "Consulting Session" and "Consultation Start Time." The former discriminates between morning and evening time slots for the consultation, while the latter provides a specific timestamp indicating the start of the consultation during the allocated session on a respective day.

3. Feature Selection Methods

Feature selection techniques occupy a critical role within the machine learning framework by systematically identifying salient variables and consequently eliminating features deemed irrelevant or redundant. In the current study, Multiple algorithms for feature selection were employed, including Pearson Correlation, Recursive Feature Elimination, Random Forest Classifier, and Reciprocal Ranking, each grounded on distinct theoretical paradigms of feature selection [20,21]. For the specific task of predicting Consultation Wait Time, the variable set was iteratively reduced until a maximum set of seven crucial features was retained. Within the framework of Pearson Correlation, the Filter Method was employed, wherein the Pearson Coefficient was utilized to quantify the linear relationship between individual features and the target variable. The resultant selected variables, ranked by their importance, encompass Consultation Start Time, Visit Time, Vitals Examination, Consulting Session, Visit Day, Cwtm5, and Priority. Contrastingly, Recursive Feature Elimination (RFE) adopts an iterative methodology that commences with the complete feature set and iteratively eliminates variables of lesser significance. Feature importance is calculated after model fitting, after which the least significant features are progressively pruned. In the current study, this feature set was determined based on a ranking derived from linear regression algorithms, yielding the selected features of Consulting Session, Visit Time, Priority, Consultation Start Time, Vitals Examination, Cwtm5, and Visit Month. Employing the Random Forest Classifier, an ensemble learning approach, a multitude of decision trees were constructed during the training phase. This algorithm offers an intrinsic attribute that quantifies feature importance based on the mean decrease in impurity across all constituent trees within the ensemble. Variables deemed most significant through this approach were Visit Time, Visit Day, Visit Month, Consultation Start Time, Cwtm5, Repeat Arrival, and Vitals Examination.

The Reciprocal Ranking methodology is an advanced feature selection technique that aggregates rankings from multiple selection algorithms to formulate an integrated ranking system. Notably, it has demonstrated superior performance in comparison to several state-of-the-art feature selection paradigms by enhancing feature space quality and mitigating the risk of selecting an unstable feature subset [22]. Mathematically, the rank for each item is determined by sorting the scores or evaluations in descending order. The item with the highest score is assigned rank 1, followed by the second-highest score receiving rank 2, and so on until the item with the lowest score is assigned the rank of N. The Reciprocal Rank is calculated using the equation:

$$R(f) = 1 / \sum (1 / Rj(f)) \quad (1)$$

where R(f) is the Reciprocal Rank of feature f, and Rj(f) is the rank of feature f assigned by the jth feature selection method. In the study, Reciprocal Ranking was employed to aggregate the feature rankings derived from three distinct selection methods: Pearson Correlation, Recursive Feature Elimination (RFE), and Random Forest Feature Importance. Through this amalgamation of rankings, the most crucial variables for predicting Consultation Wait Time were identified. These variables were determined to be Visit Time, Consultation Start Time, Consulting Session, Visit Day, Vitals Examination, Visit Month, and Priority.

4. Performance Measures

The study focuses on assessing the performance of predictive models for Consultation Wait Time, a multiclass classification problem encompassing three distinct categories: Class 1 (Low Waiting Time), Class 2 (Medium Waiting Time), and Class 3 (High Waiting Time). To address this complex task, the multiclass problem decomposed into three separate binary classification challenges, utilizing the One-vs-Rest method. Furthermore, the results from these binary classifiers were aggregated through the utilization of the weighted averaging technique [23]. For instance, when considering the "Low Waiting Time" class, a binary classifier was trained to differentiate between positive instances labeled as "Low Waiting Time" and negative instances formed by the combination of "Medium Waiting Time" and "High Waiting Time." Subsequently, this binary classifier could predict whether a new instance belonged to the "Low Waiting Time" class or not, providing a performance score for the positive class. To derive the final prediction for each instance, the weighted averaging technique was employed. The study evaluated the efficacy and the accuracy of the suggested approach using several metrics, including Accuracy, Precision, Sensitivity, F measure, and Area under the ROC Curve [24]. For each binary classification task, various performance measures are calculated and, a weighted average is computed. The values of the performance indicators Accuracy, Precision, Sensitivity, F measure, and Area under the Receiver Operating Characteristic curve for all models, are given in Table 2.

Table 2. Performance Measures of Models Predicting Consultation Wait Time

Performance Measures	Model 1	Model 2	Model 3	Model 4	Model 5
Accuracy	0.74	0.72	0.71	0.72	0.76
Precision	0.73	0.70	0.71	0.70	0.75
Recall	0.74	0.72	0.71	0.72	0.76
F measure	0.72	0.70	0.68	0.69	0.75
AUC	0.79	0.77	0.75	0.78	0.79

In contrast to other models, Model 5, which utilizes variables selected via Reciprocal Ranking, exhibits a significantly enhanced predictive performance. Model 5 achieves an accuracy of 0.76, a precision of 0.75, a recall of 0.76, an F-measure of 0.75, and an AUC (Area Under the Curve) score of 0.79.

5. Discussion

The study utilized a limited set of variables (from model 2 to model 5) to predict Consultation Wait Time. The Heatmap visualization generated using 'Seaborn' and 'Matplotlib' libraries in Python in Fig. 2. shows the feature comparison across models for predicting Consultation Wait Time.

The Observations are:

- Common Features: Consultation Start Time, Visit Time, and Vitals Examination are considered important by all four feature selection methods.
- Method-Specific Features: Repeat Arrival is unique to Random Forest.
- Time-related Features: Visit Time and Consultation Start Time are consistently marked as important, indicating their significant role.

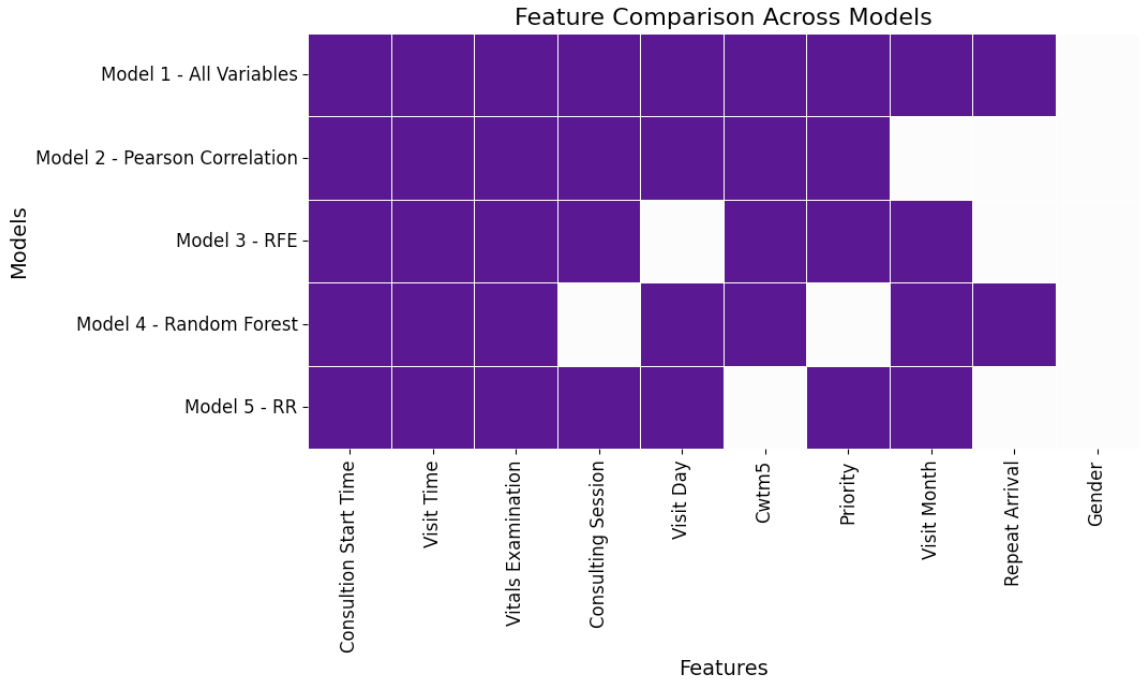


Fig. 2. Feature comparison across models for predicting Consultation Wait Time

The consistent identification of Consultation Start Time, Visit Time, and Vitals Examination as crucial factors in predicting Consultation Wait Time has significant implications for outpatient clinic management. Firstly, recognizing the importance of Consultation start time underscores the need for efficient scheduling and coordination of healthcare professionals to minimize delays and optimize patient flow. Secondly, the relevance of Visit Time highlights the significance of temporal patterns in patient arrivals, suggesting that clinics should strategically allocate resources and staff during peak and off-peak hours to mitigate wait times. Finally, the significance of the Vitals Examination emphasizes the importance of expediting initial assessments and reducing the time patients spend in pre-consultation processes. By acknowledging and acting upon these findings, outpatient clinics can enhance the overall patient experience, improve satisfaction levels, and streamline operations.

6. Limitations

The study is subject to several limitations, which are as follows. Firstly, the dataset utilized in this study was relatively small and was obtained exclusively from the general medicine outpatient department of a hospital. This limitation arises due to the challenges associated with accessing a broader range of data from the electronic health records of a hospital. Furthermore, the variables considered in the analysis were restricted to those available in the hospital database, potentially overlooking the inclusion of other relevant variables. Additionally, it is important to note that alternative machine learning techniques, such as boosting methods, were not explored for prediction purposes in this study, as the research exclusively employed neural networks. However, the study addressed the problem as a multiclass categorization issue with only three categories. However, expanding the number of categories in the prediction could yield more specific insights beneficial to patients. In the current study, the prediction model utilized associative variables and one historic trend variable to make accurate predictions. Nevertheless, it is essential to highlight that the inclusion of queueing theory variables can further enhance the performance of the prediction [15].

7. Conclusion

In summary, this study represents a significant stride in addressing a critical and widespread challenge in outpatient healthcare settings: the uncertainty in Consultation Wait Times. The research offers practical solutions that can revolutionize outpatient clinic management. By integrating the predictive models into appointment scheduling systems, clinics can provide patients with precise and reliable estimates of their wait times, ultimately leading to heightened patient satisfaction and a more positive overall experience. The innovative approach, treating wait time prediction as a multiclass classification problem and employing Reciprocal Ranking for variable selection, not only differentiates this study but also yields transformative insights into wait time prediction. Beyond enhancing the patient experience, the implementation of these findings can result in cost savings, improved resource allocation, and streamlined clinic operations. The study envisions a collaborative effort among researchers, healthcare practitioners, and policymakers to facilitate the transfer of knowledge and ensure the practical impact of this research, ultimately shaping more efficient and patient-centric outpatient healthcare delivery.

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